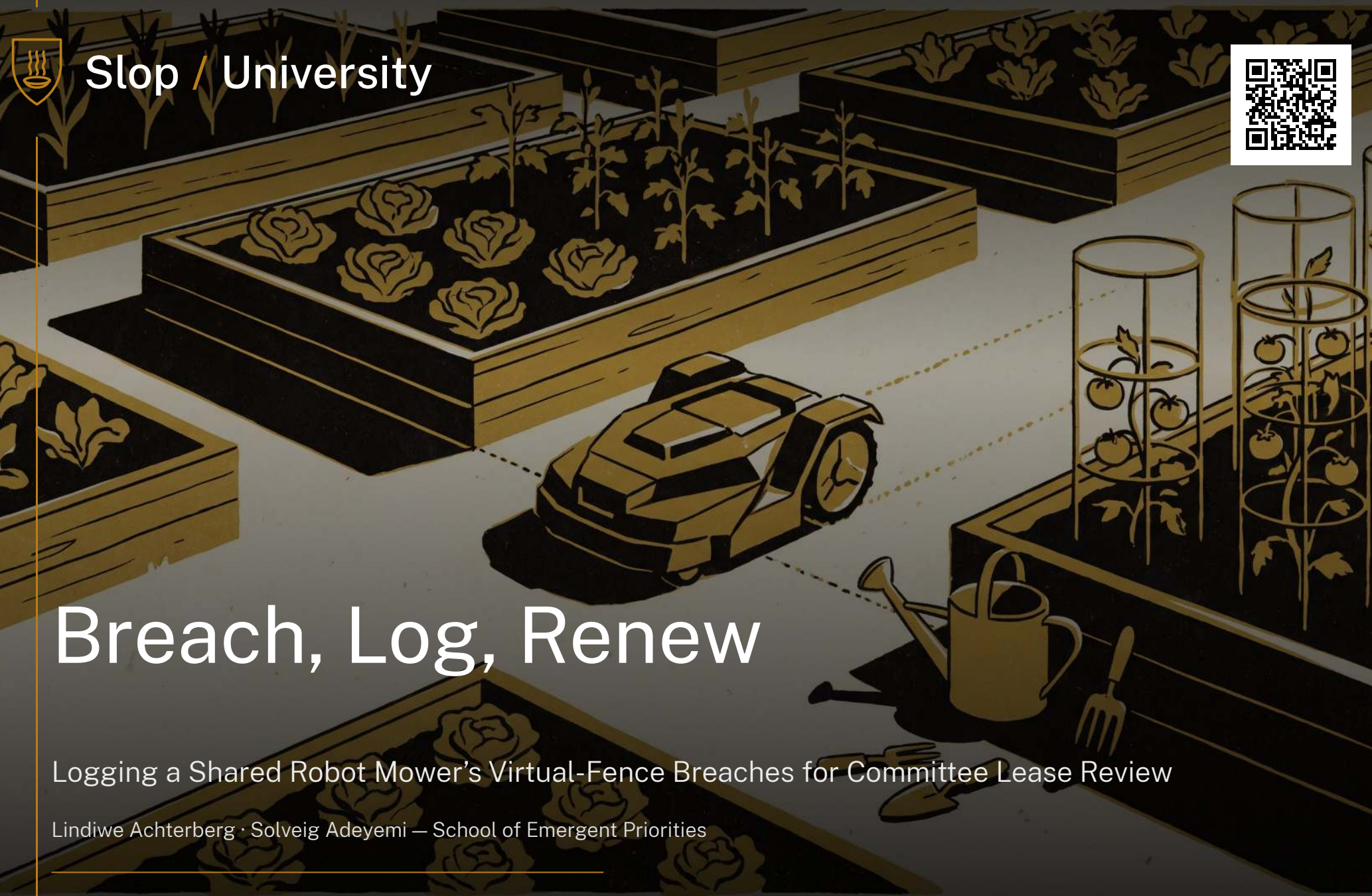


Breach, Log, Renew

Logging a Shared Robot Mower’s Virtual-Fence Breaches for Committee Lease Review

Lindiwe Achterberg · Solveig Adeyemi — School of Emergent Priorities

The problem

Shared robot mowers increasingly serve community gardens, yet virtual-fence excursions into raised beds are common and rarely traced to a cause. Little is known about whether a committee-reviewed containment ledger changes lease-renewal decisions.

Research questions

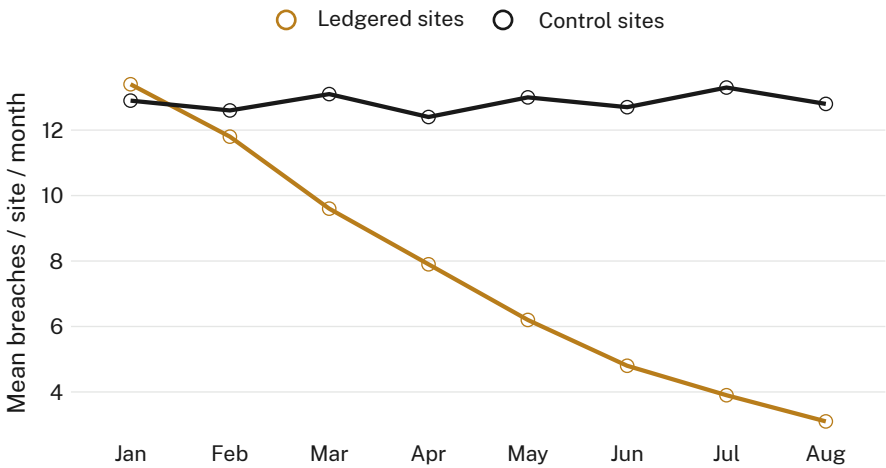
- Whether logged breach counts predict committee non-renewal at lease review
- Whether ledger visibility changes plot-holder trust in the mower
- Which virtual-fence geometries produce the most excursions per bed-hour

Approach

- 22 mower-sites (14 ledgered, 8 control) · RTK-GPS boundary telemetry · 5-min polling · 8-month deployment
- Mixed-effects model of lease non-renewal on cumulative logged breaches
- Fence geometry and firmware version held constant across sites; thresholds pre-registered

Findings

Mean monthly breaches at ledgered sites fall from double to low single figures across the deployment window. Non-renewal correlates with breach count only where the ledger is visible to the full committee ($r = 0.52$). Firmware and fence-geometry updates over the same window produce no comparable shift at the unlogged control sites.



Mean monthly virtual-fence breaches at ledgered sites fell from 13.4 to 3.1 between January and August 2025 ($n = 14$ mower-sites); 8 unlogged control sites showed no comparable decline.

Implications

Visibility, not detection, appears to drive compliance: the mower does not breach less because it is logged, but because someone is now known to be watching. A ledgered breach becomes a lease-relevant fact; an unlogged one remains gossip, and the decline tracks the ledger’s introduction rather than any change to the mower itself. The effect resembles the broader observation-changes-behaviour literature, applied here to a lawnmower rather than a ward round. Next: pre-register a non-renewal threshold before a committee sees the ledger, not after.

Selected references

1. Daniyan, I., Balogun, V., Adeodu, A., Oladapo, B., Peter, S., & Mpofu, K. (2020). Development and performance evaluation of a robot for lawn mowing. *Procedia Manufacturing*, 49, 42–48. doi:10.1016/j.promfg.2020.06.009
2. Bork, E. W., Lopes, F., Harland, K., Francis, T., Fitzsimmons, K., Carlyle, C., & Novais, C. (2025). Static evaluation of a virtual fence collar: precision and accuracy relative to a conventional GNSS collar across habitats. *Animal Biotelemetry*, 13. doi:10.1186/s40317-025-00417-1
3. Koisser, D., & Sadeghi, A.-R. (2023). Accountability of things: large-scale tamper-evident logging for smart devices. *arXiv:2308.05557*
4. McCambridge, J., Witton, J., & Elbourne, D. R. (2014). Systematic review of the Hawthorne effect: new concepts are needed to study research participation effects. *Journal of Clinical Epidemiology*, 67(3), 267–277. doi:10.1016/j.jclinepi.2013.08.015
5. Ponstingel, D. (2022). The governance of community gardens as commons and its role in the socio-ecological outcomes of gardening in Austin, Texas, USA. *Socio-Ecological Practice Research*, 4(4), 355–376. doi:10.1007/s42532-022-00133-7
6. Jacob, M., & Rocha, C. (2021). Models of governance in community gardening: administrative support fosters project longevity. *Local Environment*, 26(5), 557–574. doi:10.1080/13549839.2021.1904855
7. Chikere, I., Tolan, A., & Lindqvist, K. (2026). The Form Held Up: A Failed Replication Attempt Testing Whether a Community Garden Allotment’s Handover-Form Completeness Predicts a Successor’s Retained Horticultural Knowledge. *School of Emergent Priorities*. doi:10.5555/slop.u8gsl4

We thank the plot-holders of the instrumented gardens for scheduling access; supported by a School of Emergent Priorities seed grant.

